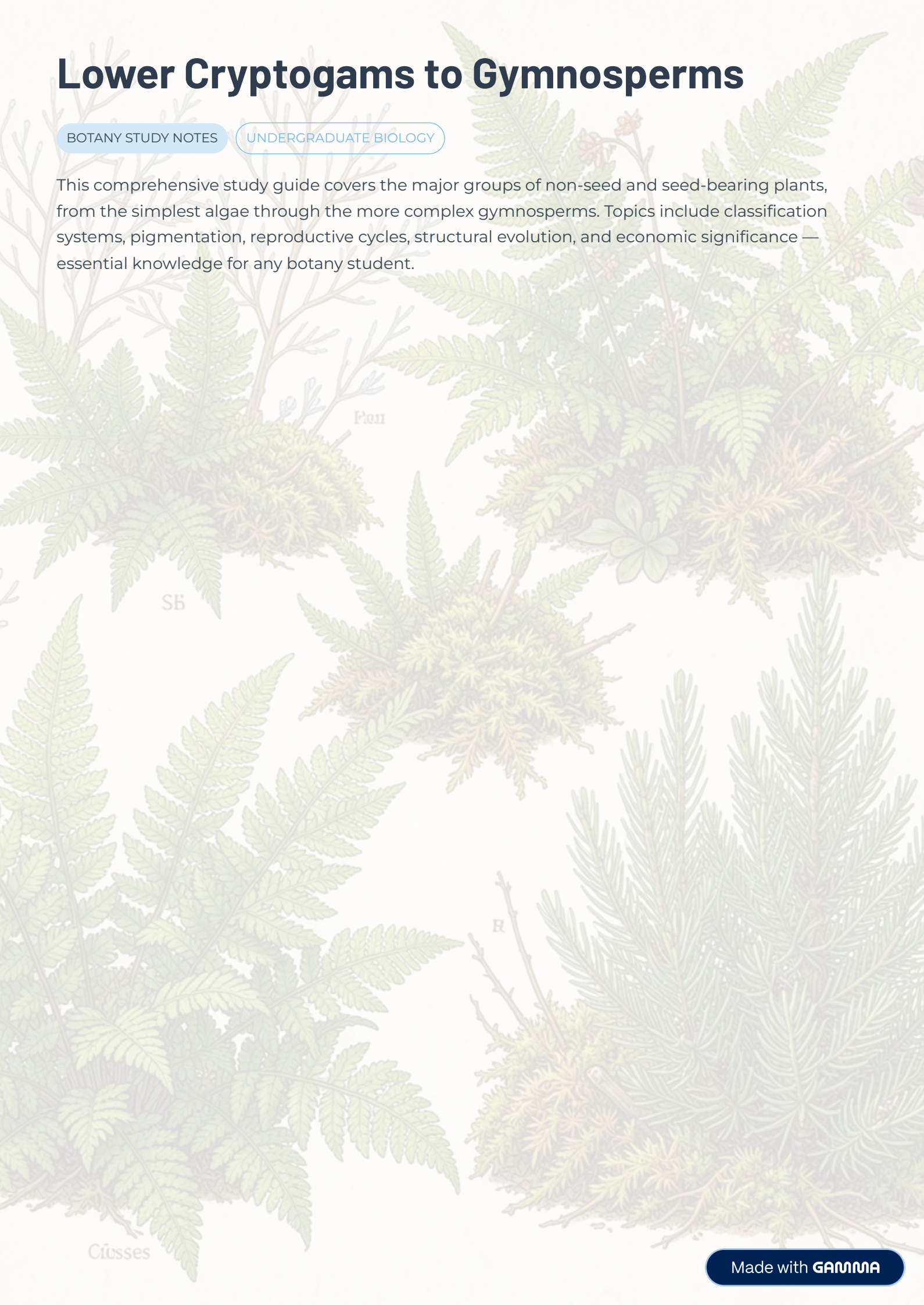


Lower Cryptogams to Gymnosperms

BOTANY STUDY NOTES

UNDERGRADUATE BIOLOGY

This comprehensive study guide covers the major groups of non-seed and seed-bearing plants, from the simplest algae through the more complex gymnosperms. Topics include classification systems, pigmentation, reproductive cycles, structural evolution, and economic significance — essential knowledge for any botany student.



Algae: Classification & Pigments

Algae are photosynthetic, primarily aquatic organisms ranging from unicellular forms to large seaweeds. They are classified mainly by their dominant pigments, stored food reserves, and cell wall composition. Unlike higher plants, algae lack true roots, stems, and leaves.

Chlorophyta – Green Algae

- Pigments: Chlorophyll a & b, carotenoids
- Stored food: Starch
- Cell wall: Cellulose
- Examples: *Chlamydomonas*, *Spirogyra*, *Ulva*

Phaeophyta – Brown Algae

- Pigments: Chlorophyll a & c, fucoxanthin
- Stored food: Laminarin, mannitol
- Cell wall: Cellulose + alginates
- Examples: *Laminaria*, *Fucus*, *Sargassum*

Rhodophyta – Red Algae

- Pigments: Chlorophyll a & d, phycoerythrin
- Stored food: Floridean starch
- Cell wall: Cellulose + agar/pectin
- Examples: *Polysiphonia*, *Gelidium*, *Gracilaria*

Cyanophyta – Blue-Green Algae

- Pigments: Chlorophyll a, phycocyanin
- Prokaryotic (no true nucleus)
- Stored food: Cyanophyceean starch
- Examples: *Nostoc*, *Anabaena*, *Oscillatoria*

Key Insight: Fucoxanthin in brown algae masks chlorophyll, giving the characteristic brown color. Phycoerythrin in red algae allows photosynthesis at greater ocean depths by absorbing blue-green light.

Algae: Life Cycles & Economic Importance

Life Cycle Types

→ Haplontic

Dominant haploid phase; zygote is the only diploid stage. Meiosis is zygotic.
Example: *Chlamydomonas*, *Spirogyra*

→ Diplontic

Dominant diploid phase; gametes are the only haploid stage. Meiosis is gametic. Example: *Fucus*

→ Haplodiplontic (Alternation of Generations)

Both phases multicellular; isomorphic or heteromorphic. Example: *Ectocarpus*, *Laminaria*, *Polysiphonia*

Economic Importance of Algae

Food & Nutrition

Porphyra (nori),
Laminaria (kombu),
Spirulina — high protein supplements

Industrial Products

Agar (*Gelidium*),
alginates (*Laminaria*),
carrageenan (*Chondrus*) — food thickening, microbiology media

Ecological Role

Produce ~50% of Earth's oxygen;
base of aquatic food webs;
nitrogen fixation by cyanobacteria

Biofuel & Biotechnology

Microalgae as biodiesel sources;
wastewater treatment; CO₂ sequestration

Fungi: Classification & Reproduction

Fungi are heterotrophic, eukaryotic organisms with chitinous cell walls. They are classified based on sexual reproductive structures and hyphal organization. Fungi play critical ecological roles as decomposers, parasites, and mutualists.



Phycomycetes (Zygomycota)

Coenocytic mycelium; asexual reproduction by sporangiospores; sexual by zygospores. Example: *Rhizopus*, *Mucor*



Ascomycetes (Sac Fungi)

Septate mycelium; sexual spores (ascospores) in asci; asexual by conidia. Example: *Aspergillus*, *Penicillium*, *Saccharomyces*



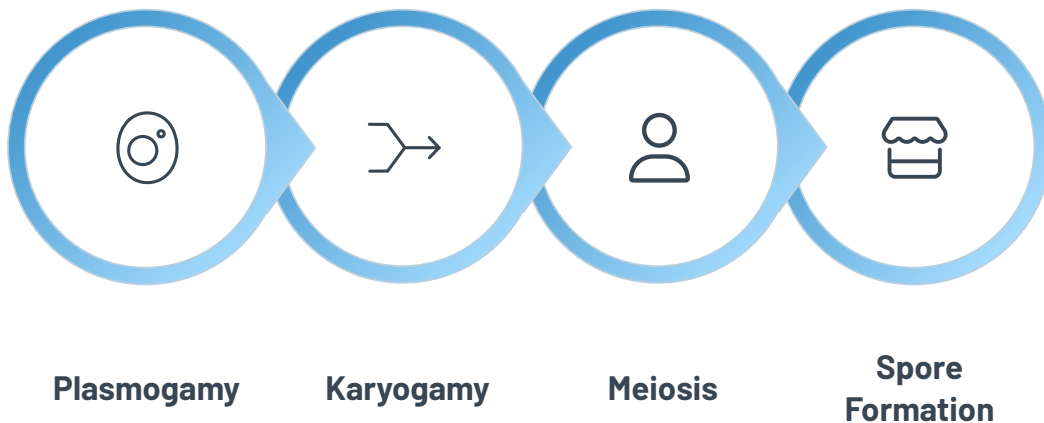
Basidiomycetes (Club Fungi)

Septate mycelium; sexual spores (basidiospores) on basidia; no asexual spores. Example: *Agaricus*, *Amanita*, *Puccinia*



Deuteromycetes (Fungi Imperfecti)

No known sexual stage; classified by conidial morphology. Example: *Alternaria*, *Cercospora*, *Helminthosporium*



Fungal reproduction involves both asexual methods (sporangiospores, conidia, oidia) and sexual methods (zygospores, ascospores, basidiospores), with the dikaryotic stage ($n+n$) being characteristic of Ascomycetes and Basidiomycetes.

Fungi: Plant Diseases & Mycorrhiza

Important Fungal Plant Diseases

Disease	Pathogen	Host
White Rust	<i>Albugo candida</i>	Crucifers
Loose Smut	<i>Ustilago nuda</i>	Wheat
Black Stem Rust	<i>Puccinia graminis</i>	Wheat
Downy Mildew	<i>Plasmopara viticola</i>	Grapevine
Red Rot	<i>Colletotrichum falcatum</i>	Sugarcane
Early Blight	<i>Alternaria solani</i>	Potato
Wilt	<i>Fusarium oxysporum</i>	Tomato, Cotton

⚠️ **Heteroecious Rust:** *Puccinia graminis* requires two hosts — wheat (primary) and barberry (alternate) — to complete its life cycle.

Mycorrhiza: Fungal-Plant Symbiosis

Mycorrhiza is a mutualistic association between fungal hyphae and plant roots, enhancing nutrient uptake — especially phosphorus — in exchange for carbohydrates.

Ectomycorrhiza

Fungal sheath (mantle) around roots; Hartig net between cortical cells. Example: *Amanita*, *Boletus* with *Pinus*

Endomycorrhiza (VAM)

Fungal hyphae penetrate root cells; form arbuscules and vesicles. Example: *Glomus* species; most common type

Significance: Increases drought tolerance, pathogen resistance, and soil aggregation; essential in sustainable agriculture and forest ecology.

Lichens: Types & Economic Importance

Lichens are symbiotic associations between a fungus (mycobiont, usually Ascomycete) and an alga or cyanobacterium (photobiont). The fungus provides structure and protection; the photobiont provides food through photosynthesis.



Crustose Lichens

Crust-like, tightly adhered to substrate; cannot be removed without damage. Examples: *Rhizocarpon*, *Graphis*, *Lecanora*



Foliose Lichens

Leaf-like lobes attached by rhizines; loosely attached to substrate. Examples: *Parmelia*, *Physcia*, *Peltigera*



Fruticose Lichens

Shrubby, branched, three-dimensional; attached at a single point. Examples: *Usnea*, *Cladonia*, *Ramalina*



Industrial Uses

Litmus dye (*Roccella*); perfumes (*Evernia*); antibiotics (usnic acid from *Usnea*)



Biological Indicators

Highly sensitive to air pollution (SO₂); used as bioindicators of air quality and environmental health



Food & Fodder

Cladonia rangiferina (reindeer moss) — primary winter food for reindeer; *Umbilicaria* eaten in Japan



Soil Formation

Pioneer organisms on bare rocks; secrete acids that weather rock and initiate primary succession

Bryophytes: Life Cycle & Evolutionary Significance

Bryophyte Life Cycle

Bryophytes exhibit a **haplodiplontic life cycle** with a dominant, independent gametophyte generation. The sporophyte is dependent on the gametophyte.

01

Spore Germination

Haploid spores germinate to form protonema

02

Gametophyte Development

Protonema develops into leafy gametophyte with antheridia and archegonia

03

Fertilization

Flagellated sperm swims to egg; requires water; forms diploid zygote

04

Sporophyte Formation

Zygote → embryo → sporophyte (foot, seta, capsule); meiosis → spores

Evolutionary Significance

First Land Plants

Bryophytes represent the earliest successful colonization of land by plants (~470 mya); bridge between algae and vascular plants

Key Adaptations

Cuticle, stomata (in some), multicellular gametangia (antheridia/archegonia), and embryo retention — precursors to seed plant traits

Limitations

No true vascular tissue; no true roots; water-dependent fertilization; dominant haploid phase — primitive characteristics

Ecological Role

Soil formation, moisture retention, nutrient cycling; *Sphagnum* forms peat bogs — important carbon sinks

Pteridophytes: Classification, Stelar Evolution & Heterospory

Pteridophytes are seedless vascular plants with true roots, stems, and leaves. They dominated Carboniferous forests and include ferns, horsetails, and clubmosses.

Psilopsida

Most primitive; no true roots or leaves.
Example: *Psilotum*, *Tmesipteris*

Lycopsida

Microphylls; strobili present. Example: *Lycopodium*, *Selaginella*

Sphenopsida

Jointed stems; reduced leaves in whorls.
Example: *Equisetum*

Pteropsida (Filicopsida)

True ferns; large fronds (megaphylls).
Example: *Dryopteris*, *Pteris*, *Adiantum*

Stelar Evolution

The stele (central vascular cylinder) evolved from simple to complex forms:

1

Protostele

Solid vascular core; no pith. Most primitive. Example: *Psilotum*, *Lycopodium*

2

Siphonostele

Hollow cylinder with pith; ectophloic or amphiphloic. Example: *Equisetum*

3

Dictyostele

Siphonostele broken into vascular bundles (meristeles). Example: Ferns

Heterospory

Production of two types of spores — **microspores** (male) and **megaspores** (female) — is a key evolutionary advance. It led to seed habit. Example: *Selaginella*, *Salvinia*

Gymnosperms: Cycas, Pinus & Gnetum

Gymnosperms are naked-seeded plants with exposed ovules. They dominated Mesozoic landscapes and include conifers, cycads, and gnetophytes.



Cycas

Living fossil;
dioecious; pinnate leaves; coralloid roots with N₂-fixing cyanobacteria; motile sperm (largest in plant kingdom); ovules on megasporophylls



Pinus

Most economically important gymnosperm; dimorphic shoots (long & dwarf); needle leaves in fascicles; winged pollen; seeds in woody cones; mycorrhizal roots



Gnetum

Most advanced gymnosperm; vessel elements in xylem (angiosperm-like); broad leaves; double fertilization-like process; connects gymnosperms to angiosperms evolutionarily

Fossil Pteridophytes

Rhynia

Silurian-Devonian; simplest vascular plant; protostele; no leaves or roots

Lepidodendron

Carboniferous tree lycophyte; up to 40m tall; contributed to coal formation

Calamites

Carboniferous tree sphenophyte; ancestor of *Equisetum*; jointed stems

Psilophyton

Devonian; dichotomous branching; terminal sporangia; early vascular tissue

Economic Importance & Evolutionary Significance

Economic Importance of Gymnosperms

Timber & Wood

Pinus, *Cedrus*, *Sequoia* — construction, furniture, paper pulp (softwood)

Resins & Oils

Turpentine, rosin, tar from *Pinus*; Canada balsam (*Abies*) for microscopy

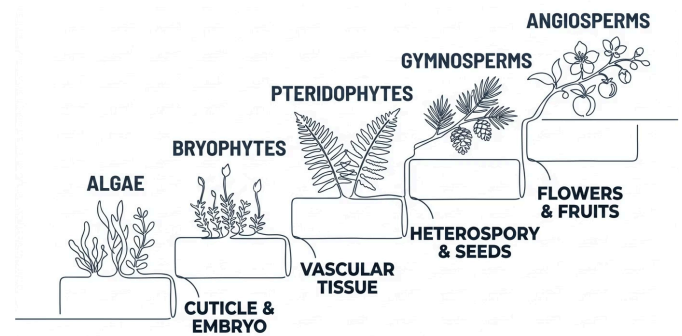
Food & Medicine

Pine nuts (*Pinus gerardiana*); *Ephedra* — ephedrine for asthma; *Taxol* from *Taxus* — anticancer drug

Ornamental & Ecological

Cycas, *Juniperus* — landscaping; soil conservation; carbon sequestration

Evolutionary Significance – Summary



The transition from lower cryptogams to gymnosperms represents major evolutionary innovations: vascular tissue, true roots and leaves, heterospory, pollen tubes, and seeds — each adaptation reducing dependence on water and enabling colonization of drier habitats.

- ✓ **Key Takeaway:** Heterospory in pteridophytes (e.g., *Selaginella*) and the seed habit in gymnosperms represent the critical evolutionary bridge to flowering plants.